

as2_personal_expense_tracker.py

```
1
2 def load_expenses(filename):
3     expenses = []
4
5     try:
6         with open(filename, "r") as file:
7             for line in file:
8                 line = line.strip()
9
10                if not line:
11                    continue
12
13                parts = line.split(",")
14
15                if len(parts) != 2:
16                    print("Skipping bad line:", line) # optional debug
17                    continue
18
19                category = parts[0].strip()
20                amount = float(parts[1].strip())
21
22                expenses.append((category, amount))
23
24    except FileNotFoundError:
25        return []
26
27    return expenses
28
29 #add expense
30 def add_expense(filename, category, amount):
31     if amount <= 0:
32         raise ValueError("Amount can't be negative ")
33
34     with open(filename, "a") as file:
35         file.write(f'{category}, {amount}\n')
36
37
38 def category_totals(expenses):
39     totals = {}
40
41     for category, amount in expenses:
42         totals[category] = totals.get(category, 0) + amount
43
44     print('\n ----Category Total----')
45     print('Category          Total')
46
47     grand_total = 0
48
49     for category, amount in totals.items():
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50         print(f'{category:<15}: ${amount:.2f}')
51         grand_total += amount
52
53     print("-----")
54     print(f'Grand Total      : ${grand_total:.2f}')
55
56     return totals
57
58
59 # list comprehension
60 # n = [1,2,3]
61 # result=[]
62 # for i in n:
63 #     result.append(i**2)
64
65 # print(result)
66
67 # n= [1,2,3]
68 # result=[i**2 for i in n]
69 # print(result)
70
71 # for c, a in expenses:
72 #     if a> limit:
73 #         return category, amount
74
75 def above_threshold(expenses, limit): #def funName(argument1, argument2 )
76     return [ (category,amount) for category,amount in expenses if amount > limit]
77
78
79 filename = r'dlytica\expenses.txt' #r'filepath' => raw string
80
81 expenses = load_expenses(filename) #load data from file
82
83 category_totals(expenses) #generate report
84
85 limit = 100
86
87 print('\nExpenses above $100:')
88 for category, amount in above_threshold(expenses, limit):
89     print(f'{category:<15}: $ {amount:.2f}')
90
91
92

```